

Group Project 1: Having a BLAST - OUTLINE

Objectives of the outline

- To check-in on the progress of writing and testing programs
- To implement constructive peer review
- To help you improve the quality of your project before it is ultimately graded

Suggested elements of your outline

In a written document, ultimately submitted as a pdf, outline your groups current status, progress made so far and plans to complete your project (included who has done what and who will do what). I have copied the project outline at the end of this assignment as a reminder. The exact format of the outline is yours to decide as a group - this outline should be a useful roadmap for your group so choose whatever format is the most helpful to you. I suggest you touch base as a time and include in your outline status updates for the following elements:

1. CS elements

- does each group member understand the N-W algorithm?
- how will you implement scoring and path matrices?
- what is the pseudocode for `countMatrices` and `longest_consecutive`?

2. Bio elements

- what test data have you decided to use?
- what conclusions are you expecting to be able to make - what hypothesis might you be testing and what results do you expect?

3. Reflections

- how will you distribute the work of the write-up?
- Note: individual peer reviews will be submitted as part of the outline check-in but will be individual (in a different gradescope submission) so they are confidential

Reminder of Your Mission

You are to write a program that takes two protein sequences as input (hard coded is ok) and outputs the best alignment using the Needleman-Wunsch algorithm. You will get the most out of this project if you put effort into working as a team: communicating with your colleagues early and often. Additionally, in your write-up you should answer the following questions:

1. Is your program doing a global or a local alignment?
2. Based on your answer to the previous question, explain why you got the test results you got.
3. In what ways could you modify the alignment matcher you wrote to be more biologically sound?

Reminder of the rubric

Please turn in all Python files, including `nw_blast.py`, and the write-up named `nw_blast.pdf` on grade-scope. Don't forget to put the honor code affirmation in the comments of both.

	Exceeding expectations (1x)	Arriving at expectations (1.5x)	Not meeting Expectations (0.5x)
CS elements (6 pts)	Correct implementation of N-W algorithm	Correct logic of N-W algorithm but with semantic error	Incorrect logic of N-W algorithm
	Correct scoring and path matrices (including boundary conditions)	Scoring and path matrices incorrect for boundary conditions	Incorrect scoring and path matrices
	Correct <code>countMatches</code> and <code>longest_consecutive</code> functions	Correct <code>countMatches</code> xor <code>longest_consecutive</code> functions	Incorrect <code>countMatches</code> and <code>longest_consecutive</code> functions
Bio elements (6 pts)	Appropriate test data	Some appropriate test data	Insufficient appropriate test data
	Conclusions regarding biological soundness	Conclusions partially regarding biological soundness	Conclusions not regarding biological soundness
Reflective (8 pts)	Thorough and understandable write-up of results	Thorough xor understandable write-up of results	Neither thorough nor understandable write-up of results
	Thorough and reflective write-up of team dynamics (peer review)	Thorough xor reflective write-up of team dynamics (peer review)	Neither thorough nor reflective peer reviews